

Project ID:

J26-DS-329

1. Topic (12 words max)

Smart Vanilla Plantation Management Using AI and Digital Twins

2. Research group the project belongs to

CoEAI - Centre of Excellence for AI

3. Specialization of the project belongs to

Data Science (DS)

4. If a continuation of a previous project:

Project ID	
Year	

5. Brief description of the research problem including references (300 words max) – references not included in word count.

Nature/Gravity of the Problem

Vanilla cultivation at the Galagedara plantation operated by Saponi dal Mondo PVT (Ltd), Kandy District, Sri Lanka, consists of approximately 2000 plants distributed across 19 blocks and four cultivation zones. Plantation supervisors, field officers, and managers currently rely on manual record-keeping practices to document plant observations, environmental conditions, and maintenance activities. This approach leads to delayed access to information, fragmented historical records, transcription errors, and difficulties in tracking disease outbreaks and plantation-wide growth patterns. Consequently, stakeholders face challenges in making timely, accurate, and data-driven decisions related to plantation management and crop health monitoring.

Technical Impact: The absence of a centralized digital infrastructure limits data accessibility, automated monitoring, predictive analytics, and AI-driven decision support capabilities. Modern artificial intelligence and computer vision techniques have demonstrated significant potential for agricultural monitoring and disease detection, yet the plantation's current workflow restricts their practical adoption [1], [2].

Economic Impact: Delayed disease identification, inefficient resource allocation, and inconsistent monitoring practices can negatively affect crop productivity and increase operational costs. Since vanilla is a high-value export crop, effective plantation management directly influences profitability and long-term business sustainability [3].

Environmental Impact: Decisions related to irrigation, fertilizer application, and disease management are primarily based on experience rather than data-driven insights, potentially leading to inefficient resource utilization and environmental degradation. Precision agriculture technologies have been shown to support more sustainable resource management practices [4][5].

Social Impact: Plantation staff spend considerable time collecting, managing, and interpreting paper-based records, increasing workload and reducing operational efficiency. A digitalized management system can improve accessibility to information and support more efficient decision-making processes.

FoC Research Cluster and Industry Vertical

Research Cluster: Centre of Excellence in Artificial Intelligence (CoEAI)

Industry Verticals: AgriTech, Smart Agriculture, Precision Agriculture, Plantation Management, and Artificial Intelligence.

Sustainable Development Goals (SDGs)

This research contributes to SDG 2 (Zero Hunger) by improving agricultural productivity and crop management through intelligent monitoring and decision support. It supports SDG 9 (Industry, Innovation and Infrastructure) through the adoption of AI-driven digital technologies and smart agricultural systems. Furthermore, it contributes to SDG 15 (Life on Land) by promoting sustainable resource utilization, improved disease management, and environmentally responsible cultivation practices through data-driven decision-making [4][5]

6. Brief description of Existing Research and Systems (300 words max)

Summary of Existing Products/Systems

- Plantix (PEAT GmbH): AI-powered mobile app for disease detection in crops. Supports 60+ crops but does not cover *Vanilla planifolia* specifically. No digital twin or GPS proximity features. No growth forecasting module.
- AgriApp / CropIn: Farm management platforms focused on Indian agriculture. Provide field mapping and basic analytics. Not deployable without internet and do not include computer vision vine measurement.

Summary of Existing Research

- Mohanty et al. (2016) - Plant Village CNN achieved 99.35% accuracy on disease detection but used controlled laboratory images, not field-condition plantation photos. Accuracy drops significantly under real-world lighting and background variation (Barbedo, 2018).
- Fuentes et al. (2017) - Deep learning for tomato disease detection in field conditions using faster R-CNN. Demonstrated feasibility of field-deployed CNN but did not address vine crops or GPS integration.
- Kamilaris & Prenafeta-Boldú (2018) - Reviewed 40 deep learning studies in agriculture; noted the scarcity of datasets for specialty/exotic crops including vanilla.
- Chen et al. (2014) - XGBoost introduced; widely applied to agricultural yield and growth forecasting with strong performance on small structured datasets.

Gaps / Limitations Identified

- Limited research exists on AI-based disease detection for *Vanilla planifolia* using field-collected plantation images, as most studies focus on common agricultural crops and controlled datasets [1][2][3][6].
- Existing agricultural research primarily addresses individual tasks such as disease detection, monitoring, or forecasting, with limited integration into a comprehensive plantation intelligence platform [1][5].
- Current Digital Twin implementations in agriculture are often application-specific and do not integrate disease detection, growth monitoring, forecasting, and recommendation generation within a unified system [4].
- Most smart agriculture platforms rely on continuous internet connectivity, limiting their applicability in plantation environments with unstable network access [4][5].
- Automated vine length measurement using computer vision remains underexplored, particularly for climbing crops such as vanilla [1][6].
- Existing Explainable AI research mainly focuses on improving transparency in disease classification rather than providing context-aware decision support for plantation management and recommendations [6].

7. Brief description of the solution's nature, including a conceptual diagram (maximum 500 words).

The proposed **AI-Powered Vanilla Plantation Management System** is an integrated intelligent platform designed to support data-driven decision-making in *Vanilla planifolia* cultivation. The solution combines computer vision, machine learning, digital twin technology, and AI-assisted decision support to improve plantation monitoring, disease management, growth assessment, and operational planning.

System Perspective

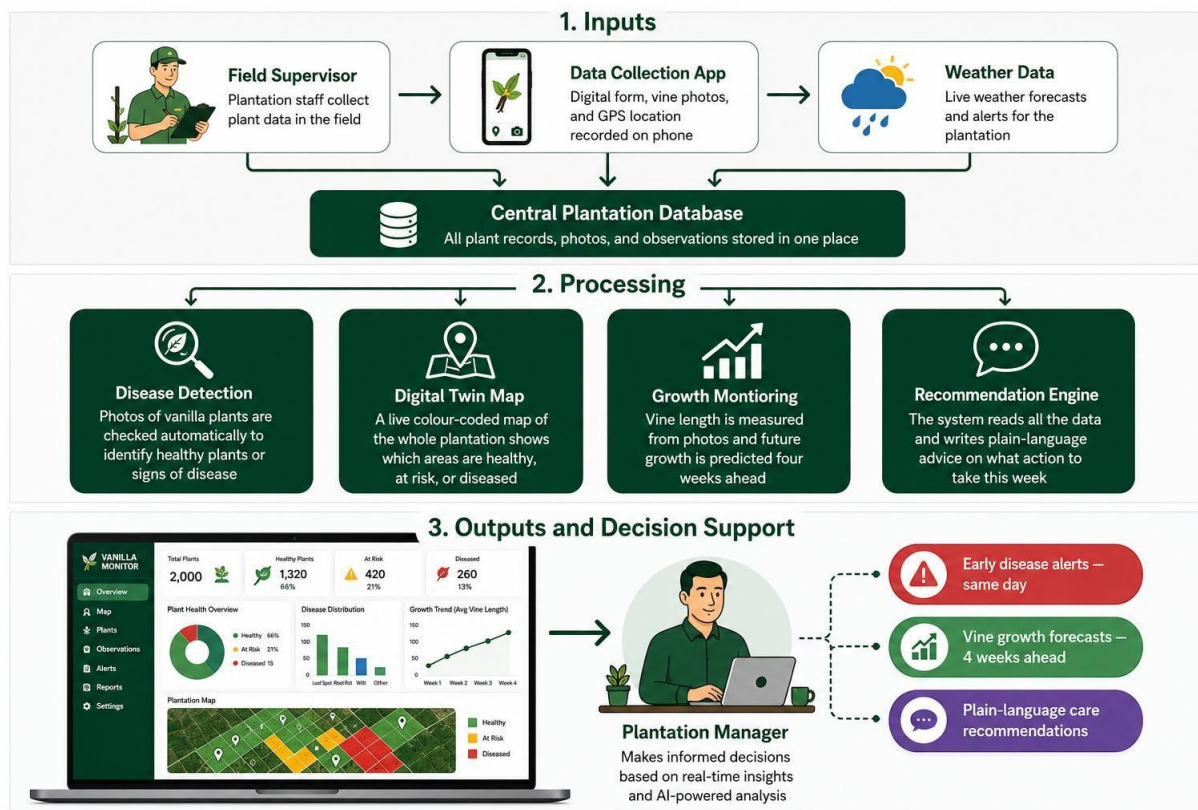
Input: Digital plantation inspection records, vanilla plant photographs, GPS coordinates, environmental observations, and weather information collected from plantation operations.

Process: The collected data are analyzed through four integrated intelligent modules: (1) disease detection using deep learning-based image analysis, (2) a GPS-enabled digital twin for spatial plantation monitoring, (3) computer vision and machine learning models for growth measurement and forecasting, and (4) an AI-assisted recommendation engine for decision support.

Output: Early disease detection alerts, real-time plantation health visualization, growth monitoring and forecasting insights, and context-aware management recommendations that support timely and informed plantation management decisions

Four AI Modules

- **Module 1 - Disease Detection:** A deep learning-based image classification module that analyzes vanilla plant images to identify healthy plants and common disease conditions. The module supports early disease detection and provides confidence-based predictions to assist plantation supervisors in timely intervention.
- **Module 2 - Digital Twin:** A GPS-enabled digital twin that provides a real-time spatial representation of plantation conditions. Supervisors can access location-aware plant information, health indicators, and block-level status through an interactive plantation map, improving monitoring and field decision-making.
- **Module 3 - Growth Monitoring & Forecasting:** A computer vision module automatically measures vine growth from plantation photographs and tracks changes over time. Machine learning techniques are used to forecast future growth trends, enabling proactive management of underperforming plants and plantation zones.
- **Module 4 - Intelligent Recommendation Engine:** An AI-assisted decision support module that analyzes plantation observations, disease status, growth patterns, and environmental conditions to generate actionable recommendations. The module provides context-aware guidance to support cultivation planning, maintenance activities, and disease management decisions.



8. Brief overview of the availability of necessary specialized domain expertise, knowledge, and data. (500 words max)

Domain Expertise

- Vanilla cultivation domain expertise is provided by the plantation supervisor Mr. U.K.Kodikara (Sapori dal Mondo PVT Ltd, Galagedara), who will validate health labels, agronomic recommendations, and disease identification training data. An independent agronomist will co-evaluate NLP recommendation quality as part of the evaluation protocol (target: $\geq 80\%$ of recommendations rated $\geq 4/5$ for actionability).
- The research team has access to the complete plant inspection forms for all 19 blocks as historical reference. The plantation is active and accessible for field visits, GPS coordinate recording, and orange reference marker installation on support trees.

Technical Knowledge Acquisition Plan

- Team members possess prior knowledge of Python, machine learning, and web application development. Additional expertise in deep learning and transfer learning will be acquired through TensorFlow/Keras documentation, online tutorials, and research literature.
- Computer vision techniques for image analysis and vine measurement will be developed using OpenCV and related image-processing resources, with validation against manual field measurements.
- Forecasting models will be developed using established machine learning frameworks and supported by literature on predictive analytics and agricultural forecasting.
- AI-assisted recommendation generation will be implemented using publicly available large language model APIs and NLP frameworks. Relevant prompt engineering and text classification techniques will be studied and evaluated during development.
- Existing experience in React-based web development will support implementation of the plantation data collection and monitoring platform.

Data Availability

- Primary data will be collected directly from the plantation through structured digital inspection forms, field observations, GPS coordinates, and vine photographs. Data collection will cover multiple blocks and cultivation zones over several months.
- Disease image datasets will be constructed using publicly available plant disease repositories, published vanilla disease image sources, and plantation photographs collected during field visits.
- Data augmentation and synthetic data generation techniques will be applied where necessary to improve class balance and model robustness while maintaining separate training, validation, and testing datasets.
- Weather and environmental information will be obtained from publicly available meteorological data sources to support growth analysis and decision-making.

Ethical Clearance

- This project does not involve human subjects, sensitive personal data, or clinical interventions. All plantation data is collected with the knowledge and consent of Sapori dal Mondo PVT (Ltd) management. Plant images and field measurements are agricultural records and do not require institutional ethical clearance beyond organizational permission, which has been granted.

9. Objectives and Novelty

To develop and evaluate an AI-powered vanilla plantation management system for Saponi dal Mondo PVT (Ltd) that integrates deep learning-based disease detection, GPS-enabled digital twin technology, computer vision-based vine growth measurement, machine learning forecasting, and NLP-driven decision support to enhance plantation monitoring and support data-driven management decisions.

Member Name with Registration No	Sub Objective	Tasks	Novelty
Holipitiya H.H.N. IT23212886	Develop an Explainable AI-based Vanilla Disease Detection System capable of accurately identifying healthy and diseased vanilla plants from field images collected under Sri Lankan plantation conditions while providing interpretable predictions to support plantation decision-making.	Collect, annotate, and validate images of healthy vanilla plants and major diseases (Anthracnose, Fusarium Stem/Root Rot, and Basal Stem Rot) from plantation field visits and relevant publicly available datasets. • Preprocess and augment the collected images to improve model robustness under varying field conditions. • Train, fine-tune, and compare state-of-the-art transfer learning models (e.g., EfficientNetV2, MobileNetV3, and ConvNeXt) for vanilla disease classification. • Apply Explainable AI techniques (e.g., Grad-CAM and Score-CAM) to visualize disease-affected regions and improve model interpretability.	Deep learning-based approaches have demonstrated strong performance in plant disease detection across many agricultural crops [2][7]. However, most existing studies rely on controlled laboratory datasets or focus on common crops, with limited research on <i>Vanilla planifolia</i> using real plantation field images [8]. Furthermore, existing disease detection models often operate as black-box systems, providing little explanation for their predictions and limiting user trust. This research proposes an Explainable AI-based transfer learning framework that detects major vanilla diseases (Anthracnose, Fusarium Stem Rot, and Basal Stem Rot) from real plantation images collected under Sri Lankan cultivation conditions while

		• Evaluate classification performance using Accuracy, Precision, Recall, F1-score, AUC-ROC, together with qualitative assessment of explanation quality. • Integrate the best-performing disease detection model into the plantation management platform to generate disease alerts.	generating visual explanations through Explainable AI techniques. The proposed framework aims to improve prediction transparency, practitioner confidence, and practical decision-making for vanilla plantation management under Sri Lankan cultivation conditions [8].
Hasapathirathna M.M.D.S.T IT23366404	Develop a GPS referenced Digital Twin of the vanilla plantation that applies spatial data analysis and machine learning to historical monthly inspection records, in order to classify plant and block-level health status, detect spatial disease clusters, and analyze health trends across blocks and zones, and to deliver these analytical outputs to field staff through location-based record retrieval.	• Compile and clean the historical plantation inspection dataset, addressing missing values, inconsistent entries, and transcription errors from the manual forms. • Geo-reference plant records by recording permanent GPS coordinates and linking each plant to its corresponding block, zone, and historical inspection records. • Perform exploratory analysis to identify distributions, seasonal variations, and relationships among plant health, environmental conditions, and plantation observations. • Prepare spatial and temporal features from historical inspection records to support plantation health analysis. • Develop predictive models to classify plant and block health status based on historical plantation observations.	Existing agricultural digital twins are typically built on IoT sensor networks and continuous cloud connectivity, and present plant data through manager-facing dashboards that require users to actively search for a plant record [9][10]. Comprehensive reviews confirm that most agricultural digital twin implementations remain dependent on dedicated sensing infrastructure [4]. However, smallholder plantations such as the Galagedara site operate without any sensor deployment, and plant condition data is recorded manually on a monthly inspection cycle, so existing sensor-driven approaches do not transfer to this setting. Furthermore, while per-plant digital twins have been demonstrated for crops such as mandarins [11], these assume reliable internet and dedicated infrastructure, and no existing research has applied a digital twin approach to Vanilla. Therefore, this

		• Analyze geo-referenced plantation records to identify disease hotspots and high-risk plantation areas. • Analyze historical inspection records to monitor health trends and changes across plants, blocks, and cultivation zones over time. • Develop a GPS-referenced digital representation of the plantation that visualizes plant, block, and zone health while allowing field staff to retrieve a plant's historical records based on their current location. • Evaluate the effectiveness of the proposed analytical framework using predictive performance, spatial analysis accuracy, historical record retrieval performance, and system usability.	research proposes a GPS-referenced digital twin constructed entirely from manually collected, low-frequency inspection records rather than sensor streams. Its contribution is analytical supervised classification models are applied to historical monthly observations to predict plant and block-level health status, spatial clustering over Haversine distance is applied to geo-referenced records to detect disease hotspots, and temporal trend analysis characterizes how block and zone health evolve across inspection cycles. Location-based retrieval then delivers a plant's analyzed historical record to field staff without manual searching, operating offline on a standard mobile device. This makes it the first data-driven, GPS-referenced digital twin for vanilla plantation management designed specifically for low-connectivity, non-instrumented smallholder conditions.
Weerasinghe W.T.C IT23213326	Develop an Automated Vine Length Measurement and Machine Learning- Based Growth Forecasting System for vanilla vine	• Collect vine growth observation records and vine photographs from plantation field visits. • Design a standardized photography protocol using a fixed reference marker for pixel-to-centimeter calibration.	Image-based techniques have demonstrated strong performance in non-destructive plant structural measurement and growth monitoring across a range of agricultural domains [12], [13]. The use of image segmentation combined with skeletonization has been validated for

	growth monitoring for four weeks ahead, using computer vision on field photographs and environmental observation data collected from the Sri Lankan plantation.	• Develop a computer vision pipeline to segment and skeletonize vine images for total vine length measurement. • Validate vine length measurements against manual tape measurements across all plantation zones. • Train and compare machine learning models for vine height forecasting. • Engineer time-series features from plantation observation data and train machine learning models for 4-week vine height forecasting. • Evaluate all models on real held-out plantation data and integrate outputs into the plantation management dashboard.	extracting the centerline path of biological objects and computing their length directly from photographs, including in fish length estimation [20] and leaf area measurement [21]. Reference-scale calibration methods - where a physical object of known size is visible in the image - have been shown to reliably convert pixel measurements into real-world dimensions under outdoor field conditions [14]. Semantic image segmentation approaches have further been applied for vine phenotyping in vineyard environments, enabling automated extraction of structural plant traits [22]. However, no existing research addresses automated length measurement of winding tropical vines such as <i>Vanilla planifolia</i>, which traverse support structures multiple times and cannot be captured by simple height estimation - a gap that is unique to vanilla cultivation. Furthermore, no published work combines image-based vine length measurement with machine learning forecasting models trained on real outdoor plantation data in Sri Lanka [15]. Therefore, this research
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			proposes a system that automates vine length measurement from field photographs and forecasts future vine growth to support data-driven plantation management under Sri Lankan vanilla cultivation conditions.
Fonseka D.W.J.M. IT23212954	Develop a Sinhala and Sinhala-English code-mixed query understanding approach that interprets farmer queries while considering relevant plantation context and domain knowledge to generate adaptive vanilla plantation management recommendations.	• Gather domain-specific vanilla cultivation knowledge, including cultivation guidelines, and other relevant vanilla agricultural practices. • Develop a Sinhala and Sinhala-English code-mixed query understanding approach to interpret farmer queries to identify relevant information for recommendation generation. • Implement a retrieval-augmented knowledge base for vanilla cultivation guidelines and domain-specific decision support. • Generate adaptive vanilla plantation management recommendations by considering the domain knowledge and relevant plantation context. • Evaluate the effectiveness of the developed solution using appropriate evaluation methods.	Existing agricultural recommendation systems have primarily focused on applying machine learning models, rule-based decision support systems, and data-driven scoring approaches to provide recommendations on crop selection, irrigation scheduling, fertilization planning, and disease management based on limited environmental parameters. [16][17][18]. However, existing agricultural recommendation approaches have limited support for Sinhala and Sinhala-English code-mixed farmer queries, creating a significant gap that limits the ability of such systems to effectively interpret local farmers' queries in their naturally used language. In particular, limited research has explored query understanding approaches specifically designed to process Sinhala and Sinhala-English code-mixed agricultural queries for recommendation generation. Furthermore, many existing systems

			provide generalized recommendations without adequately considering plantation-specific context. Moreover, few studies have investigated intelligent recommendation approaches specifically designed for vanilla cultivation under the unique environmental and cultivation conditions of Sri Lankan plantations. [19]. Therefore, this research proposes a Sinhala and Sinhala-English code-mixed query understanding approach that interprets farmer queries and considers relevant plantation context and domain knowledge to support the generation of adaptive vanilla plantation management recommendations under Sri Lankan vanilla cultivation conditions.
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References

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10. This section is to be completed by the
Supervisor and the Co-supervisor of the Project.

a) This research topic possesses a comprehensive scope suitable for a final-year project.

Yes	☒	No	☐
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b) The proposed topic exhibits novelty.

Yes	☒	No	☐
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c) The student group can successfully execute the proposed project.

Yes	☒	No	☐
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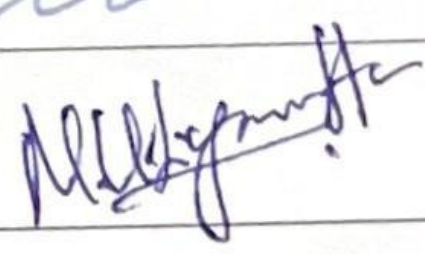
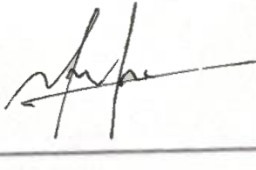
d) The proposed sub-objectives reflect the students' areas of specialization.

Yes	☒	No	☐
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e) Data required for the project is available.

Yes	☒	No	☐
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f) Any other comments:

	Title	First Name	Last Name	Signature
Supervisor*	Dr.	Mahima	Weerasinghe	
Co-Supervisor*	Dr.	Mahesh	Liganawatta	
External Supervisor	Mr.	Upali	Kodikara	

Summary of external supervisor's (if any) experience and expertise
Possesses practical experience and expertise in vanilla cultivation, plantation supervision, disease identification, and field-based agricultural management practices.

*- At this stage, identifying the Co-Supervisor is compulsory. Students should reach out to a staff member, with guidance from their supervisor, to serve as the co-supervisor for this project.

This part is to be filled by the Topic Screening Panel members.

Project ID: J26-DS-329

Acceptable: Mark/Select as necessary

Topic Assessment Accepted	
Topic Assessment Accepted with minor changes (should be followed up by the supervisor)*	✓
Topic Assessment to be Resubmitted with major changes*	
Topic Assessment Rejected. Topic must be changed	

* Detailed comments given below

Comments

Conduct a good literature survey to identify the appropriate approaches and techniques to be used.
 Confirm the datasets for the project.
 → Highlight individual research contribution properly.

The Review Panel Details

Member's Name	Signature	Date
Dr. Lakmini Abeywardhana		7/7/26
Thamali Dassanayake		7/7/26

*Important:

1. According to the comments given by the panel, make the necessary modifications and get approval from the Supervisor or the Same Panel.
2. If the project topic is rejected, identify a new topic, and follow the same procedure until the topic is approved by the assessment panel.